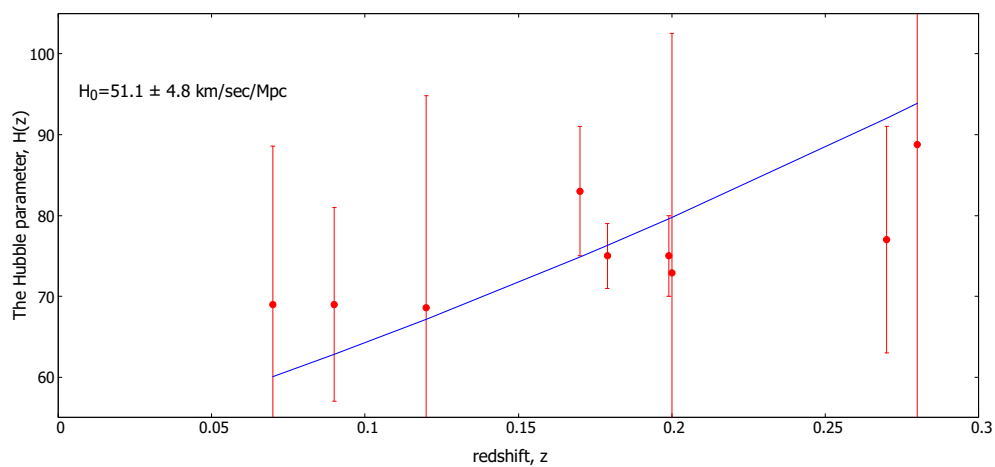


Graphical Abstract

New velocity-dependent relativistic corrections of the order of $\alpha^2 \frac{v^2}{c^2}$ to spectral lines of moving galaxies change strongly their redshifts and the Hubble constant in the nearby Universe

A.I. Agafonov



Highlights

New velocity-dependent relativistic corrections of the order of $\alpha^2 \frac{v^2}{c^2}$ to spectral lines of moving galaxies change strongly their redshifts and the Hubble constant in the nearby Universe

A.I. Agafonov

- The velocity-induced corrections to spectral lines of moving atoms follow from special theory of relativity
- These relativistic corrections for the hydrogen spectral lines $2P_{3/2,1/2} \rightarrow 1S_{1/2}$ were calculated
- The redshift and the Hubble parameter are determined both the Doppler effect and the relativistic corrections
- The present-epoch Hubble constant $H_0 = 51.2 \pm 11.0$ km/s/Mpc is significantly smaller than those published

New velocity-dependent relativistic corrections of the order of $\alpha^2 \frac{v^2}{c^2}$ to spectral lines of moving galaxies change strongly their redshifts and the Hubble constant in the nearby Universe

A.I. Agafonov^{a,b,*}

^aNational Research Center "Kurchatov Institute", Kurchatov sq. 1, Moscow, 123182, Russian Federation

^bMoscow Aviation Institute (National Research University), Volokolamskoe Shosse 4, 125993, Moscow, Russian Federation

ARTICLE INFO

Keywords:

galaxy spectral lines
velocity-dependent relativistic corrections to the spectral lines
the Hubble constant in the nearby Universe
the Hubble tension

ABSTRACT

Spectral lines of the hydrogen atoms and the alkali-like ions have been identified in the spectra of galaxies and quasars. These spectral lines are used to yield the redshifts of the objects. We show that finding the total redshift requires taking into account velocity-dependent relativistic corrections of the order of $\alpha^2 \frac{v^2}{c^2}$ to spectral lines of galaxies (α is the fine structure constant, v is the atom velocity). Although these velocity-dependent corrections follow from the energy-mass equivalence in the special relativity theory, this effect has not been previously taken into account in calculations of the redshift and, respectively, the Hubble constant. In the nearby Universe, the total redshift is determined by the combined effect of the Doppler shift and the relativistic velocity-induced shift of the spectral lines. These relativistic corrections for the lines $2P_{3/2,1/2} \rightarrow 1S_{1/2}$ of the moving hydrogen atoms were calculated. It allows us to obtain the dependence of the Hubble parameter $H(z_{obs})$ on the total redshift. Comparing the Hubble parameter with the well-known small redshift data, the present-epoch Hubble constant $H_0 = 51.2 \pm 11.0$ km/s/Mpc has been obtained. Of course, this value is significantly smaller than those presented in the well-known studies. The smaller value of H_0 is due to this new velocity-induced contribution to the redshifts of galaxies.

1. Introduction

The redshift dependence of the Hubble parameter $H(z)$ is very important for modern Cosmology (see [1, 2, 3, 4] and references therein). This dependence determines the expansion rate of the Universe down to large values $z > 1$, and may indicate a transition from decelerated to accelerated evolution of the Universe. Furthermore, measurements of $H(z)$ lead to constraints on cosmological parameters in the Standard Cosmological Model.

The present-epoch Hubble constant H_0 is widely discussed. Formally, it can be considered as the limit:

$$H_0 = \lim_{z \rightarrow 0} H(z). \quad (1)$$

The limit (1) should be calculated for small z . In this regard, intensive studies of galaxies and quasars in the nearby Universe are being conducted [5]. The edge of the nearby Universe can be estimated about 250 Mpc [6]. So, the redshifts of the objects are small, $z \simeq 0.06$, that corresponds small distances between the objects. The small distances can result in the distortion of the Hubble flow due to gravitational interactions with neighboring galaxies and their clusters. Therefore, there is a lower bound on the redshifts. The upper bond of the redshift is usually extended beyond the edge of the nearby Universe. This creates a redshift interval over which measurements are made. Respectively, the limit (1) is analyzed using the extrapolation method.

The redshift data is always used in H_0 calculations in all the known methods. It turns out that the results for H_0 obtained by different methods, differ significantly. Note that the redshift data are used in various ways. It can be the direct, model-independent calculation using the Hubble law [7, 8, 9]. The other possibility is to use the data in the model presentations of the Universe evolution, such as the $CM B + \Lambda CDM$ [10], $CM B + ACT$ [11], $CM B + SPT$ [12, 13], $DESI$ [14] and the cosmic chronometers [15, 16, 17].

*Corresponding author

 aiagafonov7@gmail.com (A.I. Agafonov)
ORCID(s): 0000-0001-8128-2086 (A.I. Agafonov)

Note that the $H(z)$ dependence implies that the Universe does not evolve with a constant rate of H_0 . So, the value of $H(z = 0)$ is ranged from 67.4 to 73.5 km/s/Mpc, while, according to data from [18], 7.8 billion years ago it was $H(z \simeq 1) = 154.0 \pm 20.0$ km/s/Mpc.

The values of H_0 obtained should be closer. Discrepancy in the Hubble constants is known as the Hubble tension [7, 8, 10, 19, 20]. Several theoretical models have been proposed to explain or resolve the problem of the Hubble tension (see [21] and reference therein). However, the proposed models cannot successfully resolve this problem (see [22] and referenced therein).

Finding the redshift z from spectral lines and its relationship with the recession velocity of galaxies and quasars, v_r , is a long-established problem. Perhaps because of this, in the many studies it did not discuss how the redshift of an object was obtained and what the spectral lines were used in calculations. Of course, the redshift is related to the emission or absorption spectrum of the galaxy. In these spectra, the spectral lines of the hydrogen atoms and the alkali-like ions C, N, O, Mg, Al, Si, Fe, Zn and others have been identified (see [23, 24, 25, 26, 27] and references therein). Analyzing the redshifts of these lines with respect to those observed on the Earth, the recessional velocities of the objects have been obtained.

In the nearby Universe, $z \ll 1$ and $v_r \ll c$, where c is the speed of light, and the usual relation between them is:

$$z = v_r/c. \quad (2)$$

The proper distance d between the observer on the Earth and astronomical objects is defined using the cosmic distance ladder. Then the Hubble constant is obtained from the equation:

$$cz = H_0 d. \quad (3)$$

According to (3), the small redshifts correspond to relatively small distances between the objects. Then the main factor that determines locations of the objects in space and directions of their movements in the nearby Universe, can be the gravitational interaction. Therefore, in the nearby Universe, the galaxy redshifts can be determined by the Doppler effect.

It should be noted that, as shown in [29], the only natural interpretation of the redshift of the nearby Galaxies is as the Doppler shift.

This differs from the cosmological redshifts which are characterized by large values $z \simeq 1$ and high. The cosmological shift is caused by an increase in the radius of curvature of space. It can be introduced under the condition that the gravitational interaction of galaxies is small compared to their kinetic energy of translational motion [28]. Then the distances between the objects must be sufficiently large in order to satisfy this condition for introducing the cosmological shift.

In present paper we argue that, at least, in nearby Universe the Eq. (3) is not correct when the redshift of a spectral line is given simply by (2). Then (3) is needed to be adjusted. The reason for this is as follows.

After the work of Salpeter and Bethe [30], the view took root that in quantum electrodynamics, as in non-relativistic mechanics, the energy spectrum of composite particles, in particular atoms and ions, does not depend on their translational motion. However, in quantum electrodynamics this statement cannot be true, since, considering the special relativity and the mass-energy equivalence, all contributions to their energy, including well-known relativistic corrections to discrete levels, determine their mass or rest energy. As atoms move, the velocity-dependent relativistic corrections shift their energy levels [31, 32]. Then the redshift due to the Doppler effect should be added by these velocity-induced shifts. This changes the equation (3) because the combined action of these two effects, namely, the Doppler shift and the velocity-induced shifts of the atomic energy levels should result in the observed redshift of the spectral lines, z_{obs} . Then Eq. (1.3) for $z \ll 1$ should be replaced by:

$$cz_{obs} = H(z)d, \quad (4)$$

where $z_{obs} = z(1 + c_1 z + c_2 z^2 + \dots)$, and the coefficients $c_{1,2}$ are numerical factors of the order of 2 for the hydrogen.

Comparing (3) with (4), we find:

$$H_0 = \frac{H(z)}{1 + c_1 z + c_2 z^2 + \dots}. \quad (5)$$

Equation (5) is the fundamental equation for finding H_0 . As shown below, the function $H(z)$ increases with z . However, with account for the definition of z_{obs} , this dependence $H(z)$ does not imply a nonlinear temporal evolution of the nearby Universe.

In this work, we calculate the coefficients $c_{1,2}$ in Eq. (5) for the lines $2P_{3/2,1/2} \rightarrow 1S_{1/2}$ of the moving hydrogen atoms. In the nearby Universe these coefficients are determined by the combined effect of the Doppler effect and the relativistic corrections of the order of $\alpha^2 \frac{v^2}{c^2}$ to the spectral lines (v is the atom's velocity). The function $H(z)$ obtained is compared with the Table data of the z -dependent Hubble parameter (see [3] and references therein). As explained in the paper, addition of the velocity-induced relativistic shifts of the atomic energy levels to the Doppler shift leads to a decreasing the magnitude of the present-epoch Hubble constant. We have found $H_0 = 51.2 \pm 11.0$ km/s/Mpc. Of course, this value is significantly smaller than those presented in the well-known studies.

2. The velocity-induced shifts of the hydrogen energy levels

Let the hydrogen atom initially be at rest. Suppose that we have calculated its energy in the state nlj ,

$$E_{nlj} = m + M + \mathcal{E}_{nlj} \quad (6)$$

where M and m are the masses of the proton and electron, respectively, and

$$\mathcal{E}_{nlj} = \langle T + V \rangle(n) + \Delta\mathcal{E}_{fine}(nj) + O_{rad}(nlj) + \dots \quad (7)$$

Here the first term on the right-hand side (7) is the sum of the average kinetic and potential energies. This contribution is proportional to α^2 and depends on the principal quantum number n . The second term is the fine structure correction, which is proportional to α^4 . This correction depends on the principal quantum number and the total angular momentum quantum number j . The last term O_{rad} is the radiative corrections (Lamb shift) plus the recoil corrections and so on. These corrections also depend on the orbital quantum number l . Using Eq. (6), the mass of hydrogen in the state nlj is:

$$m_H(nlj) = m + M + \mathcal{E}_{nlj}/c^2 \quad (8)$$

According to (8), the mass of the atom differs for different nlj states. For example, the mass in the $2P_{3/2}$ state is larger than the mass in the $2P_{1/2}$ state. This mass difference which is valid for doublet states of any ions, may play a key role in the analysis of cosmological variations of the fundamental constants [32].

Now let the atom move with a low velocity $v \ll c$. Let us analyze possible changes in the energy levels (7). The kinetic energy T in (7) contributes to the mass (8). Therefore, for the moving atom, this contribution can be multiplied by the Lorentz factor. It leads to a correction proportional to $\alpha^2 v^2/c^2$. However, the kinetic energy of the particles changes also since the intra-atomic motion of the proton and electron is supplemented by their translational motion. This leads to the appearance of relativistic perturbation operators which depend on the atom velocity. In the lowest order in the velocity, they give the corrections of order of $\alpha^n v^2/c^2$ with $n = 2, 4 \dots$ [32]. These corrections are unrelated to the Lorentz factor. The fine structure correction on the right-hand side of (7) to the atomic levels is multiplied by the Lorentz factor. As a results, we have the velocity-dependent correction of the order of $\alpha^4 v^2/c^2$.

The interaction of particles, V in Eq. (7) changes also since for moving charges, the effect of the retardation interaction between particles and the interaction through the vector potential (or magnetic interaction of moving charges) should be taken into account. These perturbation operators are also unrelated to the Lorentz factor. They leads also to the velocity-dependent corrections proportional to $\alpha^n v^2/c^2$ with $n = 2, 4 \dots$ [32].

3. The total redshift of hydrogen spectral lines

The translational motion of the atom is not completely separated from the internal motion of the electron and proton in the atom. The relativistic corrections proportional to the even powers of the atom's velocity, lead to shifts in atomic levels. In the nearby Universe the atomic velocities v are much less than the speed of light, $v \ll c$. For this reason, we omit the the velocity-induced shifts beginning with $\alpha^4 v^2/c^2$.

Considering the fine structure correction, the energy of the $2P_{1/2}$ state is (in units mc^2):

$$\epsilon_0(2P_{1/2}) = -\frac{1}{2^3}\alpha^2 - \frac{5}{2^7}\alpha^4. \quad (9)$$

For the moving atom, the energy of the $2P_{1/2}$ level becomes [32]:

$$\epsilon(2P_{1/2}) = \epsilon_0(2P_{1/2}) + \frac{11}{2^4 3}\alpha^2 \frac{v^2}{c^2}. \quad (10)$$

Similarly, for the rest atom the energy of the $2P_{3/2}$ state is given by:

$$\epsilon_0(2P_{3/2}) = -\frac{1}{2^3}\alpha^2 - \frac{1}{2^7}\alpha^4. \quad (11)$$

For the moving atom, we have:

$$\epsilon(2P_{3/2}) = \epsilon_0(2P_{3/2}) + \frac{11}{2^4 3}\alpha^2 \frac{v^2}{c^2}. \quad (12)$$

Note the correction proportional to $\alpha^2 \frac{v^2}{c^2}$ is the same for these two states because it is defined only by the principle quantum number and the orbital quantum number. The differences in the shifts occurs for the velocity-dependent fine-structure corrections of the order of $\alpha^4 \frac{v^2}{c^2}$.

For the energy of the $1S_{1/2}$ state we obtain:

$$\epsilon_0(1S_{1/2}) = -\frac{1}{2}\alpha^2 - \frac{1}{2^3}\alpha^4. \quad (13)$$

and with account of the velocity-dependent shift the energy is given by [32]:

$$\epsilon(1S_{1/2}) = \epsilon_0(1S_{1/2}) + \frac{11}{2^2 3}\alpha^2 \frac{v^2}{c^2}. \quad (14)$$

Now we can calculate the frequencies of the Lyman lines $Ly - \alpha_{1/2}$ and $Ly - \alpha_{3/2}$ of the moving hydrogen atoms. For an observer on Earth, the transition frequency is shifted by both the relativistic corrections in (10), (12) and (14), and the Doppler effect. Then the observed frequency of the transition $2P_{1/2} \rightarrow 1S_{1/2}$ is given by:

$$\omega_{obs}^{\frac{1}{2}, \frac{1}{2}} = \left(\omega_{rest}^{\frac{1}{2}, \frac{1}{2}} - \frac{11}{2^4}\alpha^2 \frac{v^2}{c^2} \right) \sqrt{\frac{1 - v_r/c}{1 + v_r/c}}. \quad (15)$$

Here v_r is the galaxy recession velocity and $\omega_{rest}^{\frac{1}{2}, \frac{1}{2}}$ is the transition frequency for the hydrogen atom at rest on Earth:

$$\omega_{rest}^{\frac{1}{2}, \frac{1}{2}} = \epsilon_0(2P_{1/2}) - \epsilon_0(1S_{1/2}) = \frac{3}{8}\alpha^2 + \frac{11}{2^7}\alpha^4 \quad (16)$$

Similar, for the transition $2P_{3/2} \rightarrow 1S_{1/2}$, we denote the frequency observed on Earth:

$$\omega_{obs}^{\frac{3}{2}, \frac{1}{2}} = \left(\omega_{rest}^{\frac{3}{2}, \frac{1}{2}} - \frac{11}{2^4}\alpha^2 \frac{v^2}{c^2} \right) \sqrt{\frac{1 - v_r/c}{1 + v_r/c}}. \quad (17)$$

where the transition frequency of the atom at rest is:

$$\omega_{rest}^{\frac{3}{2}, \frac{1}{2}} = \frac{3}{8}\alpha^2 + \frac{15}{2^7}\alpha^4 \quad (18)$$

In Eqs. (15) and (17) the two velocities describe the frequency shifts, namely the total velocity $\mathbf{v}$ and the recession velocity $\mathbf{v}_r$. In general, we would write for the total velocity:

$$\mathbf{v} = \mathbf{v}_r + \mathbf{v}_{pec}, \quad (19)$$

where $\mathbf{v}_{pec}$ is the peculiar velocity. The peculiar motions of galaxies are due to inhomogeneous matter distribution in the Universe and their gravitational interaction. The deviations from the pure Hubble flow is referred to as the peculiar velocity.

According to [33], in nearby spiral galaxies, the peculiar velocity is usually negative and is small as compared with the recession velocity v_r . However, there are rare exceptions, as shown in Table 2 of [33], when the peculiar velocities in absolute value exceeds the recession velocities. Below we use the approximation:

$$v = v_r, \quad (20)$$

that is valid for the most cases.

As is generally accepted, the redshift observed is calculated as:

$$z_{obs} = \frac{\omega_{rest}^{j, \frac{1}{2}}}{\omega_{obs}^{j, \frac{1}{2}}} - 1, \quad (21)$$

where $j = \frac{1}{2}$ for the line $2P_{1/2} \rightarrow 1S_{1/2}$, and $j = \frac{3}{2}$ for the line $2P_{3/2} \rightarrow 1S_{1/2}$.
Using (15)-(19) and (20)-(21), we have:

$$z_{obs} = \frac{1 + \frac{11}{48}\alpha^2}{\left(1 + \frac{11}{48}\alpha^2 - \frac{11}{6}\frac{v_r^2}{c^2}\right)} \sqrt{\frac{1 + v_r/c}{1 - v_r/c}} - 1 \quad (22)$$

for the line $2P_{1/2} \rightarrow 1S_{1/2}$, and

$$z_2 = \frac{1 + \frac{5}{16}\alpha^2}{\left(1 + \frac{5}{16}\alpha^2 - \frac{11}{6}\frac{v_r^2}{c^2}\right)} \sqrt{\frac{1 + v_r/c}{1 - v_r/c}} - 1. \quad (23)$$

for the line $2P_{3/2} \rightarrow 1S_{1/2}$.

With the used accuracy, the expressions (3.14) and (3.15) are approximately the same:

$$z_{obs} = \left(1 + \frac{11}{6}z^2\right) \sqrt{\frac{1+z}{1-z}} - 1, \quad (24)$$

with

$$z = \frac{v_r}{c}. \quad (25)$$

4. The Hubble parameter as a function of the redshift

Below we analyze the well-known data obtained for quasars and galaxies in nearby Universe. Here the velocities of the objects are very small as compared to the speed of light. This case may be useful for solving the Hubble tension problem.

Using (24) and (25), Eq. (4) is rewritten as:

$$\left(1 + \frac{11}{6}z^2\right) \sqrt{\frac{1+z}{1-z}} - 1 = H(z) \frac{d}{c}. \quad (26)$$

Now, considering (3) and (26), we obtain the redshift dependence of the Hubble constant:

$$H(z) = H_0 \frac{1}{z} \left[\left(1 + \frac{11}{6}z^2\right) \sqrt{\frac{1+z}{1-z}} - 1 \right]. \quad (27)$$

The dependencies (24) and (27) can be expanded in series. Up to the term z^2 , from (24) we have:

$$z_{obs} = z \left(1 + \frac{7}{3}z + \frac{7}{3}z^2\right), \quad (28)$$

and from (27) we obtain the Hubble constant:

$$H_0 = \frac{H(z)}{1 + \frac{7}{3}z + \frac{7}{3}z^2}. \quad (29)$$

Although z may be quite small, the denominator in Eq. (29) is nevertheless important. Let $z = 0.03$ and the Hubble parameter $H(z = 0.03) = 72 \text{ km/s/Mpc}$. Given the smallness of z , one might expect that, with good accuracy, the Hubble constant is $H_0 = H(z = 0.03) = 72 \text{ km/s/Mpc}$. However, this result of H_0 is greatly overestimated. Indeed, substituting $z = 0.03$ in Eq. (29), we obtain:

$$H_0 = 67 \text{ km/s/Mpc}. \quad (30)$$

The Hubble constant (30) is already close to that obtained by the Cosmic Microwave Background (CMB) measurements.

The Hubble parameter $H(z)$ for various cosmological epochs z have been presented in many papers. These results have been summarized in the common Table, often published in many articles (see, for example, Table 1 in [3] and in [36]). The results were obtained up to large redshifts, $z = 2.360$. But the dependence (29) is valid only for small redshifts. From this Table we use the nine measurements of $H(z)$ up to the value $z=0.280$. These data were obtained in the works [1, 18, 34, 35].

In Fig. 1, the nine red dots $H(z_{i=1,\dots,9})$ show these data with experimental errors. One can see a general trend of increasing the Hubble parameter with the redshift. Using (29), we describe this function $H(z)$ by the dependence:

$$H(z) = H_0 \left(1 + \frac{7}{3}z + \frac{7}{3}z^2 \right). \quad (31)$$

This dependence is shown by the blue line in Fig. 1. To find H_0 which is the present-epoch Hubble constant, we introduce the discrete function:

$$H_0(i = 1, \dots, 9) = \frac{H(z_i) \pm \Delta H(z_i)}{1 + \frac{7}{3}z_i + \frac{7}{3}z_i^2}. \quad (32)$$

Using (32), we obtain the Hubble constant H_0 :

$$H_0 = 51.2 \pm 11.0 \text{ km/s/Mpc}, \quad (33)$$

where the uncertainty refers to the confidence interval of 0.95 with the nine experimental points (the Student's coefficient of 2.31 was used).

5. Conclusion

Until now, the relativistic corrections of the order of $\alpha^2 \frac{v^2}{c^2}$ to the spectral lines of the galaxies and quasars were not taken into account. This leads to an overestimation of the objects' velocities compared to their actual ones. As a result, using Hubble's law, the overestimated values of the Hubble constant H_0 have been obtained.

The fact that these velocity-induced corrections to atomic energy levels occur, follows simply from the special theory of relativity [32].

It is shown that in nearby Universe, along with the Doppler effect, the relativistic corrections of the order of $\alpha^2 \frac{v^2}{c^2}$ to the spectral lines lead to the redshift dependence of the Hubble parameter. These relativistic corrections for the spectral lines $2P_{3/2,1/2} \rightarrow 1S_{1/2}$ of the moving hydrogen atoms were calculated. The redshift and the Hubble parameter in the nearby Universe are determined both the Doppler effect and the relativistic corrections. Respectively, the theoretical dependence of the function $H(z)$ was obtained.

The function $H(z)$ is compared with the data on the z -dependent Hubble parameter obtained for the small redshifts in [1, 18, 34, 35]. We have obtained the present-epoch Hubble constant $H_0 = 51.2 \pm 11.0 \text{ km/s/Mpc}$. Of course, this value is significantly smaller than those obtained in the well-known studies [7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17]. This is due to the fact that although these velocity-dependent corrections of the order of $\alpha^2 \frac{v^2}{c^2}$ to the spectral lines of the galaxies and quasars follow from the the energy-mass equivalence in the special relativity theory, this effect has not been previously taken into account in calculations of the redshift and, respectively, the Hubble constant.

CRedit authorship contribution statement

A.I. Agafonov is the sole author of this article. All results presented, as well as writing and editing, were performed by him.

Data availability

Data will be made available on request.

Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- [1] Simon, J., Verde, L., and Jimenez, R., 2005. Constraints on the redshift dependence of the dark energy potential. *Phys. Rev. D* 71, 123001. <https://doi.org/10.1103/PhysRevD.71.123001>
- [2] Blake, Ch., Brough, S., Colless, M., 2012. The WiggleZ Dark Energy Survey: Joint measurements of the expansion and growth history at $z < 1$. *MNRAS* 425, 405–414. <https://doi.org/10.1111/j.1365-2966.2012.21473.x>
- [3] Farooq, O., Madiyar, F.R., Crandall, S., 2017. Hubble parameter measurement constraints on the redshift of the deceleration acceleration transition, dynamical dark energy, and space curvature. *ApJ* 835, 20. doi:10.3847/1538-4357/835/1/26
- [4] Dai, X., Yang, Y., Wang, Y., Qu, Y., Yi, Sh. and Wang, F., 2026. Redshift evolution of the Hubble constant: Constraints and new insights from an interacting dark energy model. *Phys. Rev. D* 113, 063514. <https://doi.org/10.1103/zg3l-yt32>
- [5] Brout, D., Scolnic, D., Popovic, B., et al., 2022. The Pantheon+ Analysis: Cosmological Constraints. *ApJ* 938, 110. <https://doi.org/10.3847/1538-4357/ac8e04>
- [6] Böhringer, H., Chon, G., Kraan-Korteweg, R.C., Trümper, J. and Scharrel, N., 2025. Unveiling the largest structures in the nearby Universe: Discovery of the Quipu superstructure. *A&A* 695, A59. <https://doi.org/10.1051/0004-6361/202453582>
- [7] Riess, A.G., Yuan, W., Macri, L.M., et al., A Comprehensive measurement of the local value of the Hubble constant with 1 km s⁻¹ Mpc⁻¹ uncertainty from the Hubble Space Telescope and the SHOES team. *Astrophys. J. Lett.* 934, L7. <https://doi.org/10.3847/2041-8213/ac5c5b>
- [8] Freedman, W.L., Madore, B.F., Hatt, D., et al., 2019. The Carnegie-Chicago Hubble Program. VIII. An Independent Determination of the Hubble Constant Based on the Tip of the Red Giant Branch. *ApJ* 882, 34. <https://doi.org/10.3847/1538-4357/ab2f73>
- [9] Anand, G.S., Tully, R.B., Rizzi, L., Riess, A.G., and Yuan, W., 2022. Comparing Tip of the Red Giant Branch Distance Scales: An Independent Reduction of the Carnegie-Chicago Hubble Program and the Value of the Hubble Constant. *ApJ* 932, 15. <https://doi.org/10.3847/1538-4357/ac68df>
- [10] Aghanim, N., Akrami, Y., Ashdown, M. et al., 2020. Planck 2018 results VI. Cosmological parameters. *A&A* 641, A6. <https://doi.org/10.1051/0004-6361/201833910>
- [11] Aiola, S., Calabrese, E., Maurin, L., 2020. The Atacama Cosmology Telescope: DR4 maps and cosmological parameters. *JCAP* 12, (2020) <https://doi.org/10.1088/1475-7516/2020/12/047>
- [12] Balkenhol, L., Dutcher, D., Mancini, A.S., Doussot, A., Benabed, K., Galli S., et al., 2023. Measurement of the CMB temperature power spectrum and constraints on cosmology from the SPT-3G. *Phys. Rev. D* 108, 023510. <https://doi.org/10.1103/PhysRevD.108.023510>
- [13] Camphuis, E., Quan, W., Balkenhol, L., et al., 2025. CMB temperature and polarization power spectra and cosmology from 2019 and 2020 observations of the SPT-3G main field. *Phys. Rev. D* 113, 083504. <https://doi.org/10.1103/7wt3-9v2y>
- [14] Adame, A.G., Aguilar, J., Ahlen, S., et al., 2025. DESI 2024 VI: Cosmological Constraints from the Measurements of Baryon Acoustic Oscillations. *JCAP* 02, 021. <https://doi.org/10.1088/1475-7516/2025/02/021>
- [15] Hu, W. and Dodelson, Sc., 2002. Cosmic Microwave Background Anisotropies. *Annu. Rev. Astron. and Astrophys.* 40, 171-216. <https://doi.org/10.1146/annurev.astro.40.060401.093926>
- [16] Hivon, E., Gorski, K.M., Netterfield, C.B., et al., 2002. Master of the cosmic microwave background anisotropy power spectrum: a fast method for statistical analysis of large and complex cosmic microwave background data sets. *ApJ* 567, 2. <https://doi.org/10.1086/338126>
- [17] Moresco, M., 2015. Raising the bar: new constraints on the Hubble parameter with cosmic chronometers at $z > 2$. *MNRAS* 450, L16–L20. <https://doi.org/10.1093/mnras/1slv037>
- [18] Moresco, M., Cimatti, A., Jimenez, R., 2012. Improved constraints on the expansion rate of the Universe up to $z \sim 1.1$ from the spectroscopic evolution of cosmic chronometers. *JCAP* 08, 006. <https://doi.org/10.1088/1475-7516/2012/08/006>
- [19] Leizerovich, M., Landau, S.J., Scóccolac, C.G., 2024. Tensions in cosmology: A discussion of statistical tools to determine inconsistencies. *Phys. Lett. B* 855, 138844. <https://doi.org/10.1016/j.physletb.2024.138844>
- [20] Cai, R.-G., Wang, Sh.-J., 2026. The Hubble tension: A decade review. *RAA* 26, 084011. <https://doi.org/10.1088/1674-4527/ae842f>
- [21] Jia, X.D., Hu, J.P. and Wang, F.Y., 2023. Evidence of a decreasing trend for the Hubble constant. *A&A* 674, A45. <https://doi.org/10.1051/0004-6361/202346356>
- [22] Schöneberg, N., Abellán, G. F., Sánchez, A. P., et al., 2022. The H0 Olympics: A fair ranking of proposed models. *Phys. Rep.* 984, 1. <https://doi.org/10.1016/j.physrep.2022.07.001>
- [23] Savedoff, M.P., 1956. Physical Constants in Extra-Galactic Nebulae. *Nature* 178, 688. <https://doi.org/10.1038/178688b0>
- [24] Scheuer, P.A.G., 1965. A Sensitive Test for the Presence of Atomic Hydrogen in Intergalactic Space. *Nature* 207, 963. <https://doi.org/10.1038/207963a0>
- [25] Varshni, Y.P., 1989. Identification of lines in the spectrum of the quasar 0420-388. *Astrophys Space Sci* 153, 153–167. <https://doi.org/10.1007/BF00643620>
- [26] Bahcall, J.N., L. Steinhart, Ch., Schlegel, D., 2004. Does the fine-structure constant vary with cosmological epoch?. *ApJ* 600, 520. <https://doi.org/10.1086/379971>
- [27] Webb, J.K., Murphy, T.M., Flambaum, V.V., et al., 2001. Further Evidence for Cosmological Evolution of the Fine Structure Constant. *Phys. Rev. Lett.* 87, 091301. <https://doi.org/10.1103/PhysRevLett.87.091301>

- [28] L.D. Landau, E.M. Lifshitz, *The Classical Theory of Fields*, 4th ed., Pergamon, Oxford, 1975 (Paragraph 114 Redshift in Russian edition).
- [29] Bunn, E.F., Hogg D.W., 2009. The Kinematic origin of the cosmological redshift. *Am.J.Phys.* 77, 688-694. <https://doi.org/10.1119/1.3129103>
- [30] Salpeter, E.E., Bethe, H.A., 1951. A relativistic equation for bound-state problems. *Phys. Rev. A* 84, 1232. <https://doi.org/10.1103/PhysRev.84.1232>
- [31] Agafonov, A.I., 2018. Hydrogen energy-level shifts induced by the atom motion: crossover from the Lamb shifts to the motion-induced shifts. *Mod. Phys. Lett. B* 32, 1850273. <https://doi.org/10.1142/S0217984918502731>
- [32] Agafonov, A.I., 2025. The special relativistic bound-state equation for the moving hydrogen atom and velocity-induced corrections to the hydrogen energy levels. *Phys. Lett. A* 563, 131048. <https://doi.org/10.1016/j.physleta.2025.131048>
- [33] Hong, T., Staveley-Smith, L. , Masters, K.L., et al., 2019. 2MTF - VII. 2MASS Tully-Fisher survey final data release: distances for 2,062 nearby spiral galaxies. *MNRAS* 487, 2061–2069. <https://doi.org/10.1093/mnras/stz1413>
- [34] Zhang, C., Zhang, H., Yuan, S., et al., 2014. Four New Observational H(z) Data From Luminous Red Galaxies of Sloan Digital Sky Survey Data Release Seven. *RAA* 14, 1221. <https://doi.org/10.1088/1674-4527/14/10/002>
- [35] Jimenez, R., Verde, L., Treu, T., Stern, D., 2003. Constraints on the Equation of State of Dark Energy and the Hubble Constant from Stellar Ages and the Cosmic Microwave Background. *ApJ* 593, 622. <https://doi.org/10.1086/376595>
- [36] Alfano A.Ch., Cafaro, C., Capozziello, S., Luongo, Muccino, M., 2026. Investigating the cosmic distance duality relation with gamma-ray bursts. *JHEA* 49, 100444. <https://doi.org/10.1016/j.jheap.2025.100444>

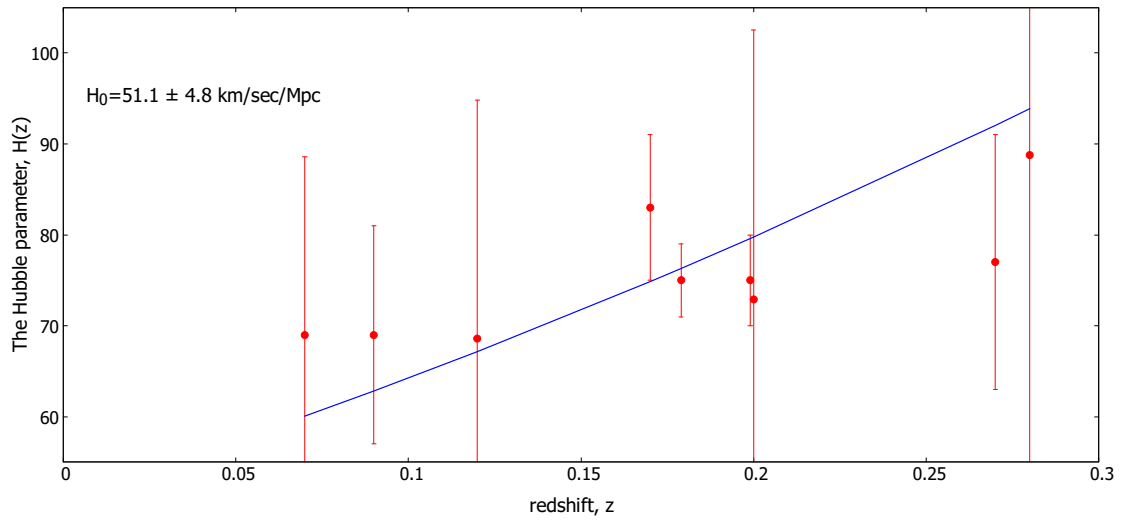


Figure 1: The Hubble parameter as a function of the redshift.